Project Title : Delivery Delay Analysis Using Power BI

Project brief on delivery data analysis

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Content

Project Objective

Problem Statement: To design an interactive, multi-page Power BI dashboard that enables analysis of delay in deliveries. Delivery times for a food delivery service are being impacted by multiple factors such as peak-hour demand and adverse weather conditions. This impacts customer satisfaction and operational efficiency. The objective is to understand how delivery time varies with factors such as city, traffic, weather, and delivery practices, and use this analysis to improve SLA compliance and optimize operations.

Original dataset: 45,584 rows × 20 columns. After cleaning: 34,487 rows.

Key fields include city (zone), time_orderd, time_taken_min, weather_conditions, traffic_density, multiple_deliveries, and festival.)

SLA (by city): Metropolitan = 27 min, Urban = 22 min, Semi-Urban = 49 min.

Delay calculation: Delay_Minutes = time_taken_min - SLA.

Colab Notebook for data cleaning , preprocessing and basic EDA with python.

Power BI as visualization tool

- **Initial Dataset Shape: 45,584 rows × 20 columns**
- **Missing Value Handling: Filled numeric columns (delivery_person_age, delivery_person_ratings) with median; categorical columns trimmed & standardized; rows with critical nulls dropped.**
- **Numeric Conversion: Converted ratings, age, multiple_deliveries to numeric type.**
- **Duplicates: Checked for duplicates (If any)**
- **Text Standardization: Lowercased, stripped spaces in columns like city, weather_conditions, road_traffic_density.**
- **Column Dropping: Removed non-essential fields (e.g., vehicle_condition, lat/long) for model simplicity.**
- **Final Shape After Cleaning: 34,487 records (from 45,584).**

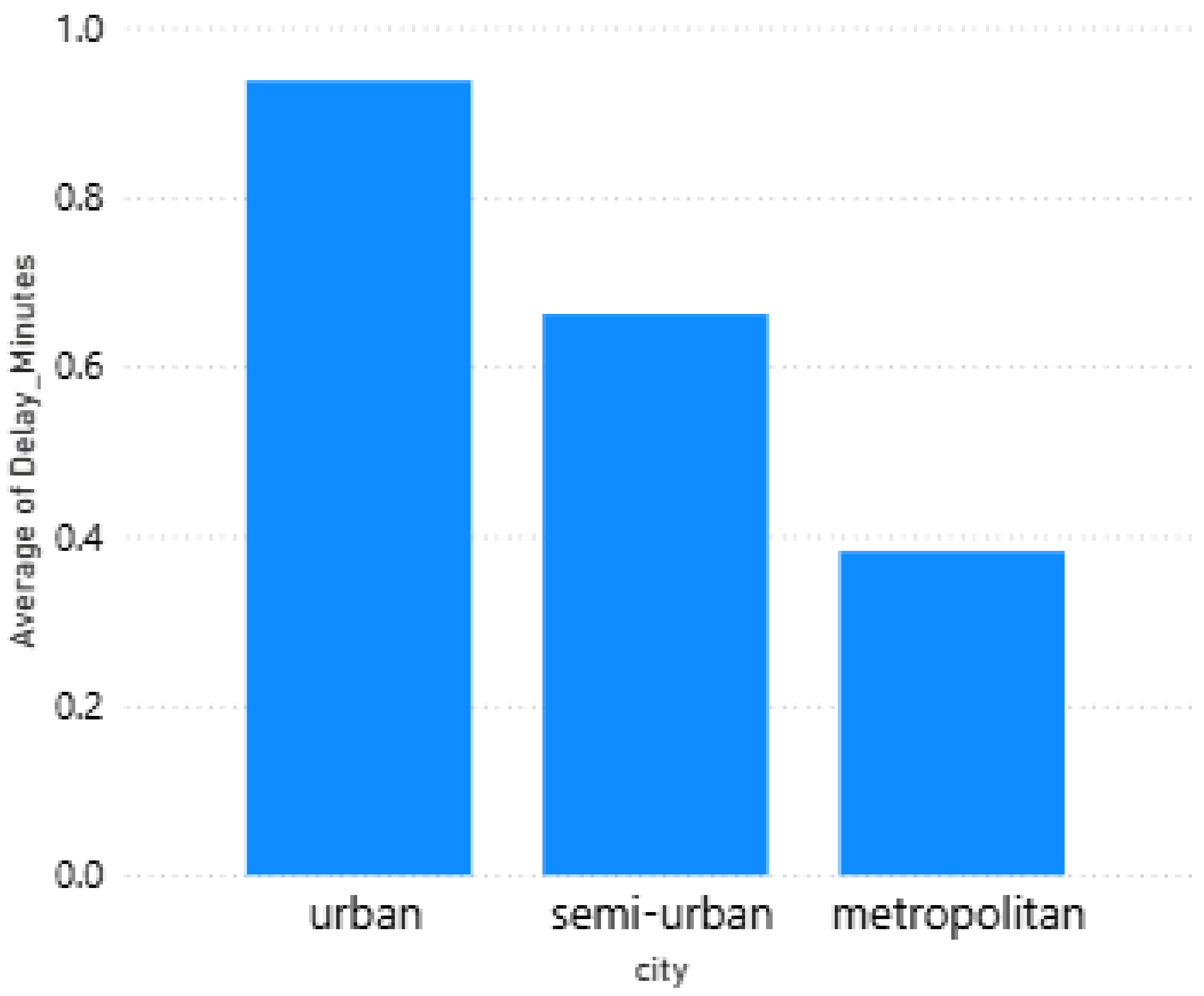
34K	45.75%	0.51
Total Orders	Delayed %	Avg Delay

delivery_person_age

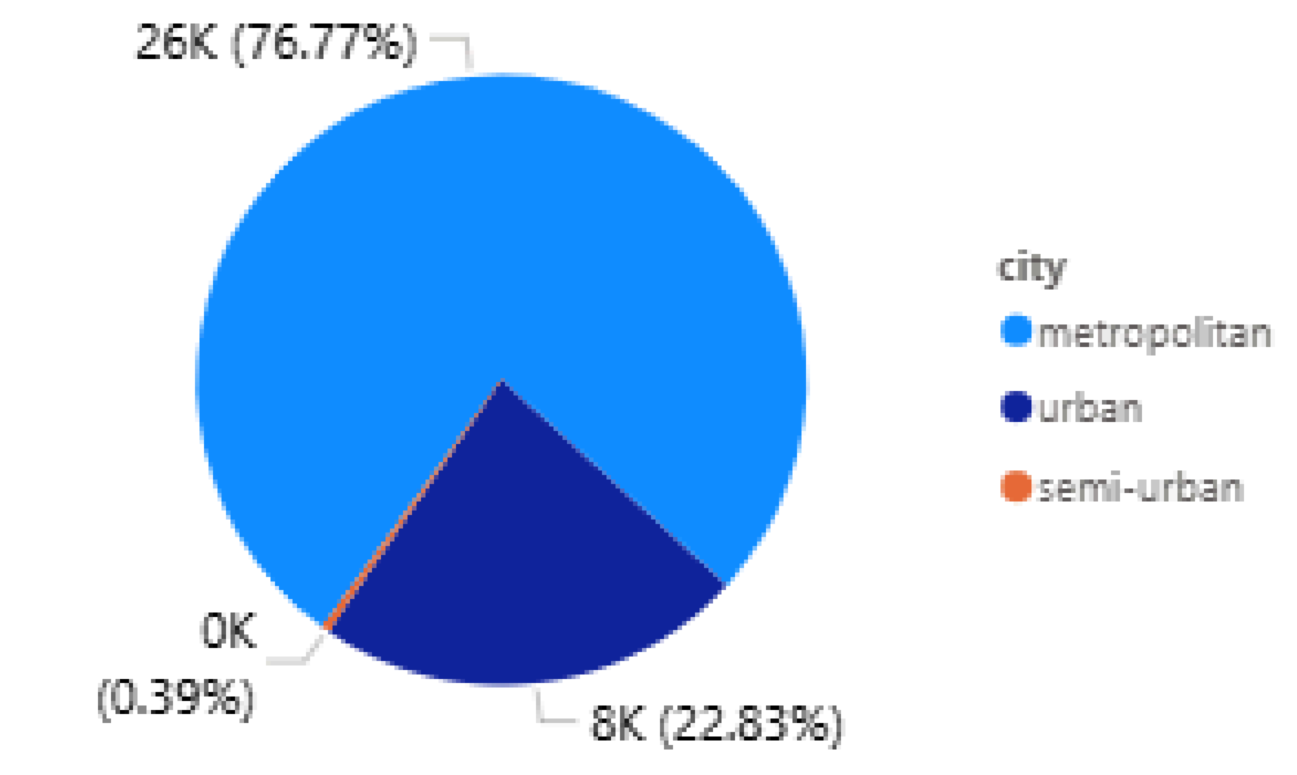
20 – 39

city	Median_SLA_By_City
metropolitan	27.00
semi-urban	49.00
urban	22.00
Total	**26.00**

Average of Delay_Minutes by city



Total Orders by city



TimeSlot: All

type_of_vehicle: All

road_traffic_density: All

multiple_deliveries: All

city: All

weather_conditions: All

festival: All

type_of_order: All

road_traffic_density: All

weather_conditions: All

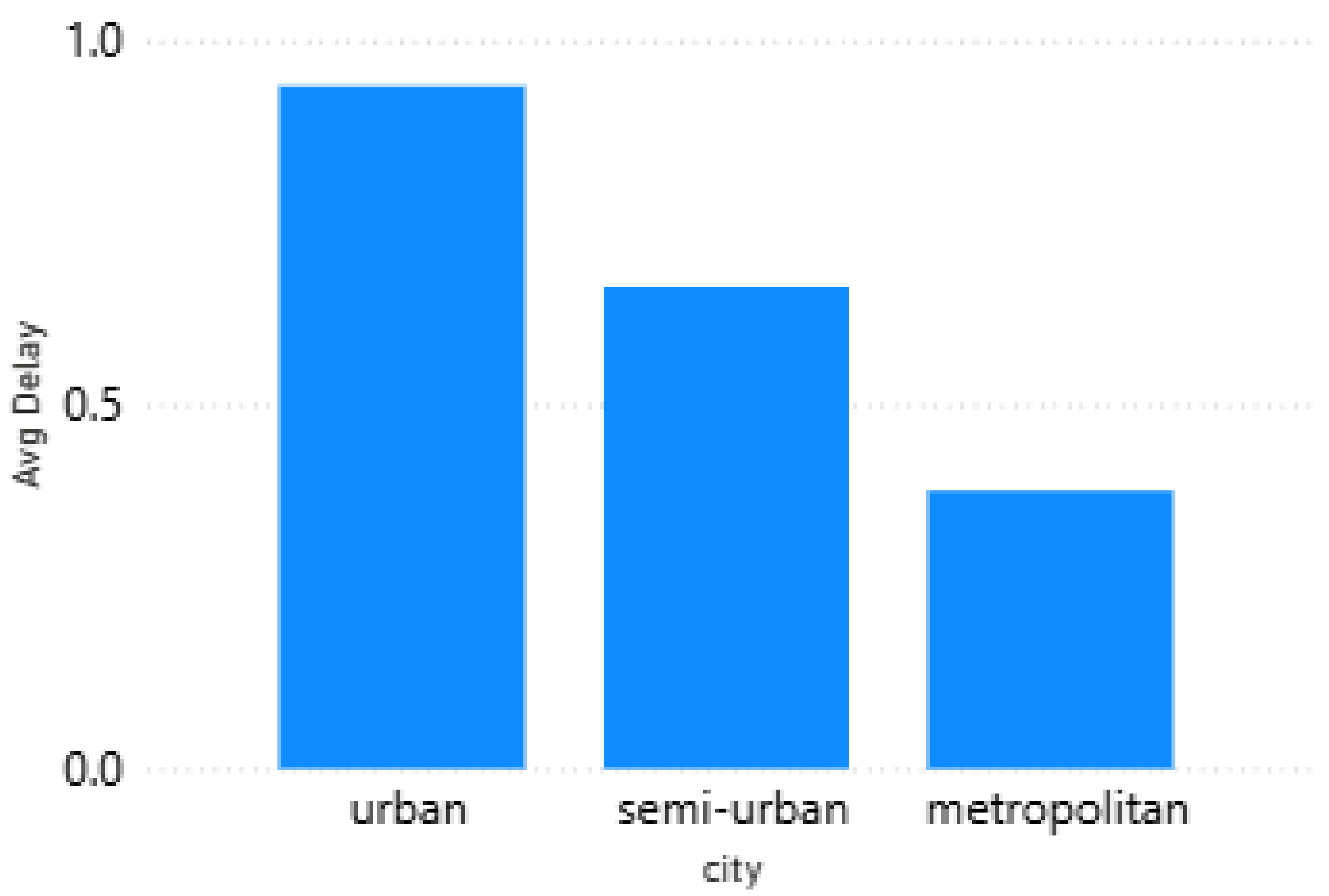
festival: All

type_of_order: All

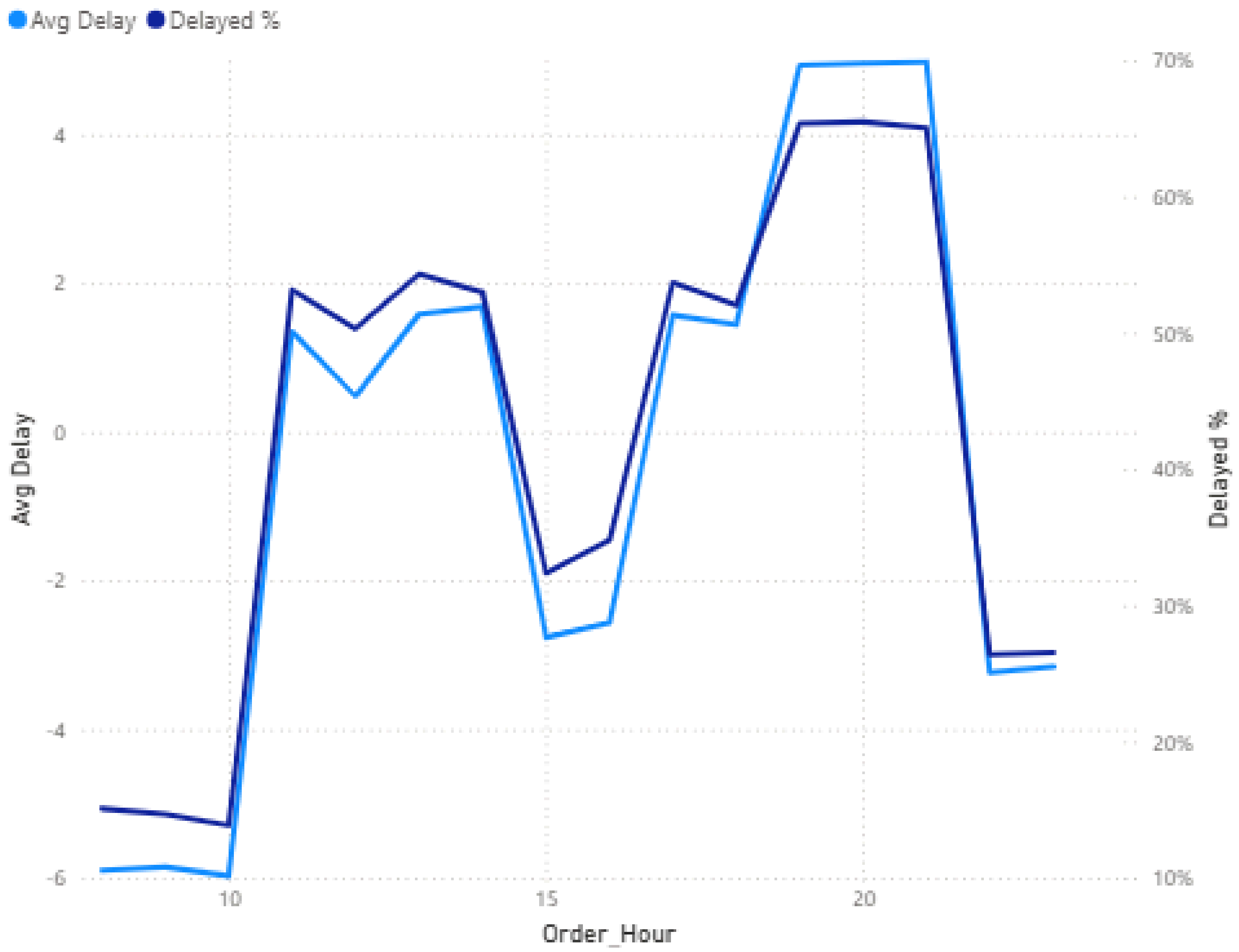
type_of_vehicle: All

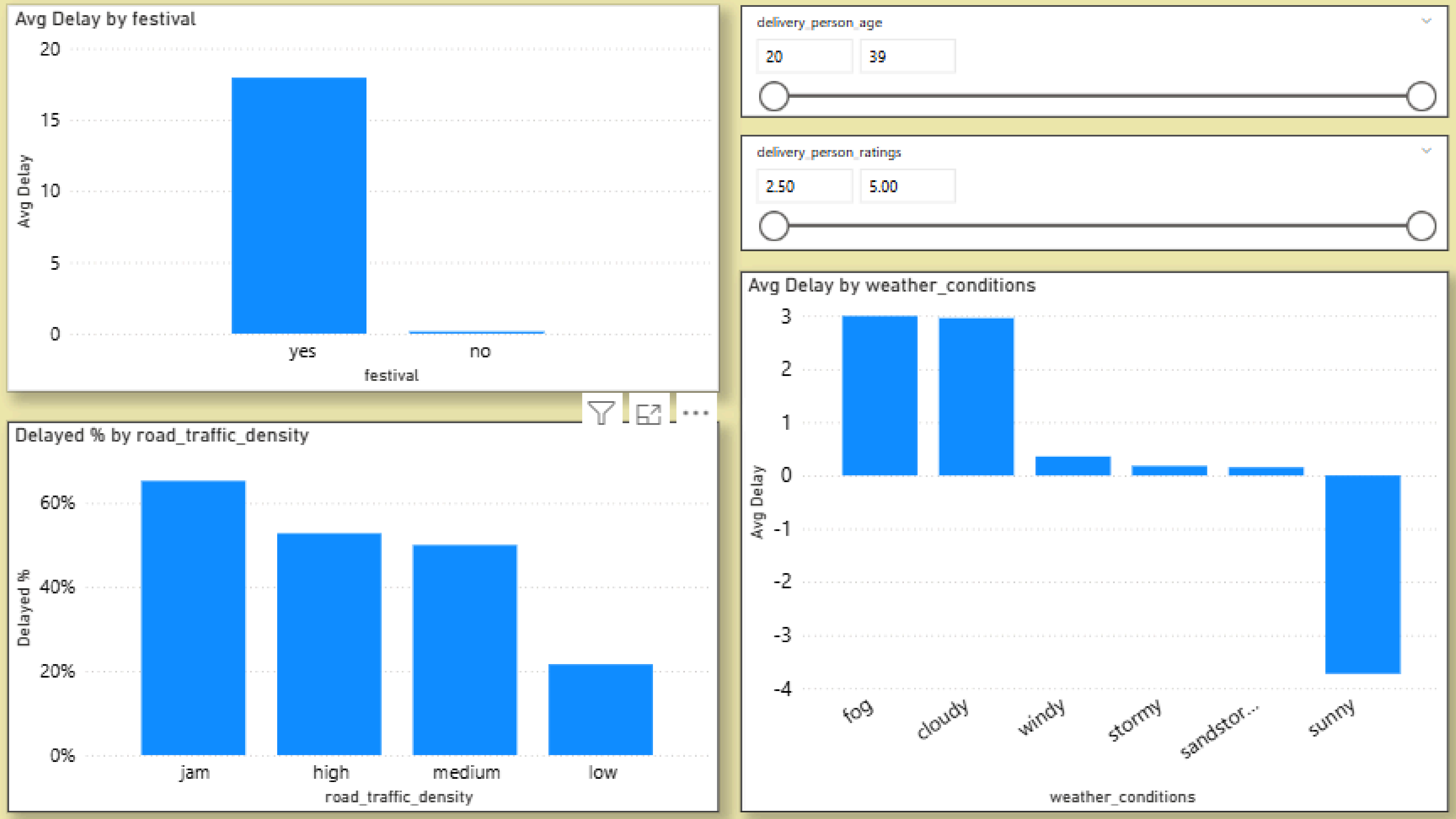
city	Afternoon	Evening	Morning	Night	**Total**
metropolitan	-0.62	3.11	-4.60	-0.50	**0.38**
semi-urban	1.33	0.48	0.75	0.92	**0.66**
urban	0.52	3.79	-2.85	0.43	**0.94**
Total	**-0.35**	**3.23**	**-4.10**	**-0.27**	**0.51**

Avg Delay by city



Avg Delay and Delayed % by Order_Hour




Avg Delay by festival
20
15
10
5
0
Avg Delay
yes
no
festival
delivery_person_age
20
39
delivery_person_ratings
2.50
5.00
Delayed % by road_traffic_density
60%
40%
20%
0%
Delayed %
jam
high
medium
low
road_traffic_density
Avg Delay by weather_conditions
3
2
1
0
-1
-2
-3
-4
Avg Delay
fog
cloudy
windy
stormy
sandstor...
sunny
weather_conditions

- Overall delayed orders: 45.7%. Avg delivery time: 26.4 min.
- Highest delay during dinner slot (19:00–21:00): Avg +5 min, ~65% late.
- Urban zone is slightly worse than Metropolitan; Semi-Urban too small for conclusion.
- Weather: Cloudy & Fog show ~+3 min delay; Sunny is favorable (-3.7 min).
- Traffic: Jam + Fog/Cloudy = critical risk (>+10 min delay, >85% late).
- Multiple deliveries (>1) cause severe overruns (up to +18 min).

- Restrict batching during dinner peak; deploy extra riders between 19:00–21:00.
- Enable weather-aware ETA adjustments during Fog/Cloudy conditions.
- Optimize routing during high traffic and micro-zone assignments to reduce travel time.
- Introduce dynamic SLAs by zone × time slot for realistic planning.

Conclusion

1. Improved understanding of delay patterns enabling better resource allocation.
2. Operational improvements leading to reduced delays and higher customer satisfaction.
3. Better preparedness for traffic/weather conditions resulting in fewer SLA breaches.

The analysis identified key delay drivers using Power BI dashboards. Metrics such as average
delay, peak delay time slots, and zone-based performance were computed. Based on findings,
actionable strategies were recommended to reduce late deliveries.